

Part-1

Laboratory 3 – Programming Basics in C++

Lab Objectives

- Online Compilers
- Work with variables and arithmetic operations (Algorithm, Flowchart, Pseudocode).
- Data Types & Comments
- Improve writing readable code using comments and identifiers.
- If statement

There are a lot of alternatives to offline IDE you can choose among them i.e.

- https://www.onlinegdb.com/online_cplusplus_compiler

- <https://www.programiz.com/cpp-programming/online-compiler/>

1. Exercise 3: Identifiers and Comments

Variables & Data Types

Int, float, char, string, bool.

```
int age = 20;  
float gpa = 3.7;  
char grade = 'A';  
string name = "Ali";  
bool isAStudent = true;
```

Rewrite Exercise 3 but follow these rules:

- Use meaningful identifiers (e.g., number1, number2, totalSum). Use camelCase for standardization
- Add comments explaining each part of your program.

Rules for Valid Identifiers

1. Must start with a letter (A–Z / a–z) or underscore `_`.

2. Can contain letters, digits (0–9), and underscores.
3. **Cannot** contain spaces or special characters like @, #, \$, %, -.
4. **Cannot** be a C++ keyword (e.g., int, while, return).

Valid Identifiers (Correct)

1. totalScore
2. num1
3. _studentName

Invalid Identifiers (Incorrect)

1. 2value → starts with a digit
2. first-name → contains a hyphen
3. my score → contains a space

2. Exercise 4: Arithmetic Operations

Write algorithm and a C++ program that takes two numbers and prints their sum, difference, and product. Additionally give simple flowchart

2.1. flowchart

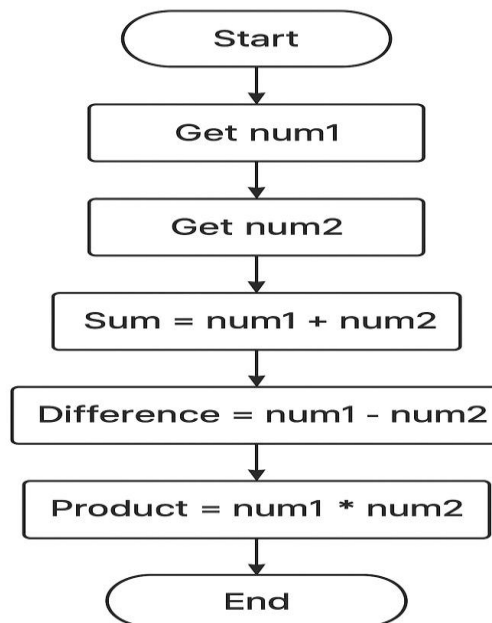


Fig.1 Simple Flowchart

2.2. Algorithm:

- Step 1: Start the program.
- Step 2: Declare two integer variables: num1 and num2.
- Step 3: Display the message: "Enter first number:"
- Step 4: Read the value and store it in num1.
- Step 5: Display the message: "Enter second number:"
- Step 6: Read the value and store it in num2.
- Step 7: Calculate the sum $\rightarrow \text{num1} + \text{num2}$.
- Step 8: Display: "Sum = " followed by the result.
- Step 9: Calculate the difference $\rightarrow \text{num1} - \text{num2}$.
- Step 10: Display: "Difference = " followed by the result.
- Step 11: Calculate the product $\rightarrow \text{num1} * \text{num2}$.
- Step 12: Display: "Product = " followed by the result.
- Step 13: End the program.

2.3. Optional Pseudocode

BEGIN

```
DECLARE num1 AS INTEGER  
DECLARE num2 AS INTEGER  
DISPLAY "Enter first number: "  
INPUT num1
```

```
DISPLAY "Enter second number: "  
INPUT num2
```

```
SET sum = num1 + num2  
DISPLAY "Sum = ", sum
```

```
SET difference = num1 - num2  
DISPLAY "Difference = ", difference
```

```
SET product = num1 * num2  
DISPLAY "Product = ", product
```

END

2.4. Program

```
#include <iostream>
using namespace std;

int main() {
    // Dynamic values
    int num1, num2;
    cout << "Enter first number: ";
    cin >> num1;
    cout << "Enter second number: ";
    cin >> num2;

    cout << "Sum = " << num1 + num2 << endl;
    cout << "Difference = " << num1 - num2 << endl;
    cout << "Product = " << num1 * num2 << endl;
    return 0;
    // Todo Add division operation
    // Reproduce with static values
}
```

3. Task

Write a program that reads three numbers and prints the smallest.

4. Questions for Review

1. What is the difference between cin and cout?
2. List three valid and invalid identifiers.

5. Quiz:

- 1- Write C++ Program to compute Area of the rectangle.
مساحة المستطيل = الطول * العرض
- 2- Write C++ Program to compute Area of the square
مساحة المربع = طول الضلع * نفسه
- 3- Write C++ program compute area or circumference of rectangle
محيط المستطيل = (الطول + العرض) * 2
مساحة المستطيل = الطول * العرض

- 4- Write C++ program to compute circumference of Square
مساحة المربع = طول الضلع * نفسه
محيط المربع = طول الضلع * 4
- 5- Write C++ program to compute the circumference of Triangle
(Optional)

Part-2

IF Condition + Comparison Programs

```
if (condition) {  
    // code to execute if condition is true  
}
```

Example of a condition:

- ✓ $A > B$
- ✓ $x == 10$
- ✓ $grade \neq 'A'$
- ✓ $age \geq 18$

New: Comparison Operations

```
#include <iostream>  
using namespace std;  
  
int main() {  
    int a = 10, b = 20;  
  
    cout << (a == b) << endl; // 0 (false)  
    cout << (a != b) << endl; // 1 (true)  
    cout << (a > b) << endl; // 0  
    cout << (a < b) << endl; // 1  
    cout << (a >= b) << endl; // 0  
    cout << (a <= b) << endl; // 1  
    return 0;  
}
```

Example 1: Check if a number is positive

```
int x;  
cout << "Enter a number: ";  
cin >> x;  
  
if (x > 0) {  
    cout << "The number is positive.";  
}
```

Syntax for if statement

important Points

- Uses *parentheses* around the condition.
- Uses *curly braces {}* to group code blocks.
- Ends each statement with a *semicolon*;
- Don't use *elif* it is for Python.

```
if (condition) {  
    // code to execute  
}  
else if (condition) {  
    // more code  
}  
else {  
    // fallback code  
}
```

Example 2: Write a program that Compare Two Numbers

```
#include <iostream>  
using namespace std;  
  
int main() {  
    int A, B;
```

```
cout << "Enter two numbers: ";  
cin >> A >> B;  
  
if (A > B)  
    cout << "A is greater than B";  
else if (A < B)  
    cout << "B is greater than A";  
else  
    cout << "Both numbers are equal";  
  
return 0;  
}
```

Example 3: Write C++ Program to check weather given number is Negative or Positive.

```
int x = 10;  
  
if (x > 0) {  
    cout << "Positive";  
}  
else if (x == 0) {  
    cout << "Zero";  
}  
else {  
    cout << "Negative";  
}
```

Example 4: Write a program that the greatest among three numbers

```
#include <iostream>  
using namespace std;  
  
int main()  
{  
    int A, B, C;  
    cout << "Enter three numbers: ";  
    cin >> A >> B >> C;
```

```
if (A > B && A > C)
    cout << "A is the greatest";
else if (B > A && B > C)
    cout << "B is the greatest";
else if (C > A && C > B)
    cout << "C is the greatest";
else
    cout << "Some numbers are equal";

return 0;
}
```

Assignment-1: Write a C++ program that find the smallest among three numbers. (اخر وقت للتسليم 2025-12-7)

```
#include <iostream>
using namespace std;
// Note we define them before main function
// write it without semicolon “;”
#define PI 3.14
#define MAX_SIZE 100

int main() {
    float radius = 5.0;
    float area = PI * radius * radius;

    cout << "PI = " << PI << endl;
    cout << "Max Size = " << MAX_SIZE << endl;
    cout << "Area of circle = " << area << endl;

    return 0;
}
// to do read radius value from keyboard
```

Logical Comparison operations

AND Operator (&&)

True **only if both** conditions are true. (And operation deal with it as product op. in other words)

A	B	A && B
true	true	true
true	false	false
false	true	false
false	false	false

OR Operator (||)

True if **at least one** condition is true.

A	B	A B
-----	-----	-----
true	true	true
true	false	true
false	true	true
false	false	false

```
#include <iostream>
using namespace std;

int main() {
    int x = 7;

    cout << (x > 5 && x < 10) << endl;
    // 1 (true)
    cout << (x == 5 || x == 7) << endl;
    // 1 (true)
    cout << !(x == 7) << endl;
    // 0 (false)

    return 0;
}
```

Source code and Lab materials could be found at the course repository.

<https://github.com/Mohammed20201991/Programming-UF-Basics>